

Java Programming Fundamentals

Student Utility Program

Experiment No: 02 **Date:** 15/09/2026 **Duration:** 2 Hours

OBJECTIVE

To understand and implement the basic programming constructs of Java, including variables, data types, operators, input/output, conditional statements, looping statements, and user-defined methods by developing a menu-driven Student Utility Program.

CONCEPTS COVERED

- Java Program Structure
- Variables and Data Types
- Operators
- Input and Output
- Conditional Statements (if, if-else, switch)
- Looping Statements (for, while, do-while)
- Methods (User-defined Functions)

TASK DESCRIPTION

Part A – Java Basics

- Display student information.
- Accept Name, Roll Number, and Marks.
- Calculate Total and Percentage.
- Display the result.

Part B – Conditional Statements

- Check whether a number is Even or Odd.
- Find the Largest of Three Numbers.
- Display Grade based on percentage using if-else.
- Display Day of the Week using switch.

Part C – Looping Statements

- Print multiplication tables.
- Display numbers from 1 to N.
- Calculate the sum of first N natural numbers.
- Generate the Fibonacci series.

Part D – Methods

- Finding the factorial of a number.
- Checking whether a number is prime.
- Finding the maximum of two numbers.

- Calculating the area of a circle.
- Every method is parameterized and returns the result.

SYSTEM REQUIREMENTS

Hardware	Standard PC/Laptop; Intel Core i3/i5 or equivalent; Minimum 4 GB RAM; 500 MB free storage
Software	Windows / Linux / macOS; JDK 17 or JDK 21; Eclipse / IntelliJ IDEA / NetBeans / VS Code; Terminal

ALGORITHM / LOGIC

1. Start the program.
2. Create a Scanner object to accept input from the user.
3. Display the main menu containing options for Parts A, B, C, D and Exit.
4. For Part A, read student name, roll number and marks, calculate total and percentage, and display the result.
5. For Part B, use if-else for even/odd and grade, comparisons for largest of three, and switch for day of week.
6. For Part C, use for, while, and do-while loops for the required looping operations and generate Fibonacci series.
7. For Part D, call parameterized methods factorial(), isPrime(), maximum(), and areaOfCircle() from main().
8. Display the returned results.
9. Repeat the menu until the user selects Exit.
10. Stop the program.

UML CLASS / PROGRAM STRUCTURE

```

StudentUtilityProgram
|
+-- main()
|
+-- factorial(int n) -> long
+-- isPrime(int n) -> boolean
+-- maximum(int a, int b) -> int
+-- areaOfCircle(double radius) -> double

```

SOURCE CODE

```

import java.util.Scanner;

public class StudentUtilityProgram {

    static long factorial(int n) {
        long fact = 1;
        for (int i = 1; i <= n; i++) {
            fact = fact * i;
        }
        return fact;
    }

    static boolean isPrime(int n) {
        if (n < 2) return false;

        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) return false;
        }
        return true;
    }

    static int maximum(int a, int b) {
        return (a > b) ? a : b;
    }

    static double areaOfCircle(double radius) {
        return Math.PI * radius * radius;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;

        do {
            System.out.println("\n===== STUDENT UTILITY PROGRAM =====");

```

```

System.out.println("1. Part A - Student Information");
System.out.println("2. Part B - Conditional Statements");
System.out.println("3. Part C - Looping Statements");
System.out.println("4. Part D - Methods");
System.out.println("5. Exit");
System.out.print("Enter your choice: ");
choice = sc.nextInt();

switch (choice) {

    case 1:
        sc.nextLine();
        System.out.println("\n--- PART A: STUDENT INFORMATION ---");

        System.out.print("Enter Student Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Roll Number: ");
        int rollNo = sc.nextInt();

        System.out.print("Enter marks of Subject 1: ");
        int m1 = sc.nextInt();
        System.out.print("Enter marks of Subject 2: ");
        int m2 = sc.nextInt();
        System.out.print("Enter marks of Subject 3: ");
        int m3 = sc.nextInt();
        System.out.print("Enter marks of Subject 4: ");
        int m4 = sc.nextInt();
        System.out.print("Enter marks of Subject 5: ");
        int m5 = sc.nextInt();

        int total = m1 + m2 + m3 + m4 + m5;
        double percentage = total / 5.0;

        System.out.println("\n--- STUDENT RESULT ---");
        System.out.println("Name          : " + name);
        System.out.println("Roll Number: " + rollNo);
        System.out.println("Total Marks: " + total);
        System.out.printf("Percentage : %.2f%%\n", percentage);

        if (percentage >= 40)
            System.out.println("Result      : PASS");
        else
            System.out.println("Result      : FAIL");
        break;

    case 2:
        int bChoice;
        do {
            System.out.println("\n--- PART B: CONDITIONAL STATEMENTS ---");
            System.out.println("1. Even or Odd");
            System.out.println("2. Largest of Three Numbers");
            System.out.println("3. Display Grade");
            System.out.println("4. Day of the Week");
            System.out.println("5. Back to Main Menu");
            System.out.print("Enter your choice: ");
            bChoice = sc.nextInt();

            switch (bChoice) {
                case 1:
                    System.out.print("Enter a number: ");
                    int num = sc.nextInt();
                    if (num % 2 == 0)
                        System.out.println(num + " is Even.");
                    else
                        System.out.println(num + " is Odd.");
                    break;

                case 2:
                    System.out.print("Enter first number: ");
                    int a = sc.nextInt();
                    System.out.print("Enter second number: ");
                    int b = sc.nextInt();
                    System.out.print("Enter third number: ");
                    int c = sc.nextInt();

                    int largest;
                    if (a >= b && a >= c)
                        largest = a;
                    else if (b >= a && b >= c)
                        largest = b;
                    else
                        largest = c;

                    System.out.println("Largest = " + largest);
                    break;

                case 3:
                    System.out.print("Enter percentage: ");
                    double p = sc.nextDouble();

                    if (p >= 90) System.out.println("Grade: A+");
                    else if (p >= 80) System.out.println("Grade: A");
                    else if (p >= 70) System.out.println("Grade: B");
            }
        } while (bChoice != 5);
}

```

```

        else if (p >= 60) System.out.println("Grade: C");
        else if (p >= 50) System.out.println("Grade: D");
        else if (p >= 40) System.out.println("Grade: E");
        else System.out.println("Grade: F");
        break;

    case 4:
        System.out.print("Enter day number (1-7): ");
        int day = sc.nextInt();

        switch (day) {
            case 1: System.out.println("Monday"); break;
            case 2: System.out.println("Tuesday"); break;
            case 3: System.out.println("Wednesday"); break;
            case 4: System.out.println("Thursday"); break;
            case 5: System.out.println("Friday"); break;
            case 6: System.out.println("Saturday"); break;
            case 7: System.out.println("Sunday"); break;
            default: System.out.println("Invalid day number.");
        }
        break;

    case 5:
        break;
    default:
        System.out.println("Invalid choice.");
    }
} while (bChoice != 5);
break;

case 3:
    int cChoice;
    do {
        System.out.println("\n--- PART C: LOOPING STATEMENTS ---");
        System.out.println("1. Multiplication Table");
        System.out.println("2. Display 1 to N");
        System.out.println("3. Sum of First N Natural Numbers");
        System.out.println("4. Fibonacci Series");
        System.out.println("5. Back to Main Menu");
        System.out.print("Enter your choice: ");
        cChoice = sc.nextInt();

        switch (cChoice) {
            case 1:
                System.out.print("Enter a number: ");
                int table = sc.nextInt();

                System.out.println("Multiplication Table of " + table);
                for (int i = 1; i <= 10; i++)
                    System.out.println(table + " x " + i + " = " + (table * i));
                break;

            case 2:
                System.out.print("Enter N: ");
                int n = sc.nextInt();
                int i = 1;

                while (i <= n) {
                    System.out.print(i + " ");
                    i++;
                }
                System.out.println();
                break;

            case 3:
                System.out.print("Enter N: ");
                int limit = sc.nextInt();
                int sum = 0, j = 1;

                do {
                    sum += j;
                    j++;
                } while (j <= limit);

                System.out.println("Sum = " + sum);
                break;

            case 4:
                System.out.print("Enter number of terms: ");
                int terms = sc.nextInt();

                int first = 0, second = 1;
                System.out.print("Fibonacci Series: ");

                for (int k = 1; k <= terms; k++) {
                    System.out.print(first + " ");
                    int next = first + second;
                    first = second;
                    second = next;
                }
                System.out.println();
                break;

            case 5:

```

```

        break;
    default:
        System.out.println("Invalid choice.");
    }
} while (cChoice != 5);
break;

case 4:
    int dChoice;
    do {
        System.out.println("\n--- PART D: USER-DEFINED METHODS ---");
        System.out.println("1. Factorial");
        System.out.println("2. Prime Check");
        System.out.println("3. Maximum of Two Numbers");
        System.out.println("4. Area of Circle");
        System.out.println("5. Back to Main Menu");
        System.out.print("Enter your choice: ");
        dChoice = sc.nextInt();

        switch (dChoice) {
            case 1:
                System.out.print("Enter a number: ");
                int factNum = sc.nextInt();
                System.out.println("Factorial = " + factorial(factNum));
                break;

            case 2:
                System.out.print("Enter a number: ");
                int primeNum = sc.nextInt();

                if (isPrime(primeNum))
                    System.out.println(primeNum + " is Prime.");
                else
                    System.out.println(primeNum + " is Not Prime.");
                break;

            case 3:
                System.out.print("Enter first number: ");
                int x = sc.nextInt();
                System.out.print("Enter second number: ");
                int y = sc.nextInt();

                System.out.println("Maximum = " + maximum(x, y));
                break;

            case 4:
                System.out.print("Enter radius: ");
                double radius = sc.nextDouble();

                System.out.printf("Area of Circle = %.2f%n",
                    areaOfCircle(radius));
                break;

            case 5:
                break;
            default:
                System.out.println("Invalid choice.");
        }
    } while (dChoice != 5);
    break;

case 5:
    System.out.println("\nThank you for using Student Utility Program.");
    break;

default:
    System.out.println("Invalid choice.");
}

} while (choice != 5);

sc.close();
}
}

```

COMPILATION & EXECUTION

Save as: StudentUtilityProgram.java

Compile:
javac StudentUtilityProgram.java

Run:
java StudentUtilityProgram

SAMPLE OUTPUT SCREEN

```
===== STUDENT UTILITY PROGRAM =====
1. Part A - Student Information
2. Part B - Conditional Statements
3. Part C - Looping Statements
4. Part D - Methods
5. Exit
Enter your choice: 1

--- PART A: STUDENT INFORMATION ---
Enter Student Name: Rajesh
Enter Roll Number: 101
Enter marks of Subject 1: 85
Enter marks of Subject 2: 78
Enter marks of Subject 3: 92
Enter marks of Subject 4: 88
Enter marks of Subject 5: 80

--- STUDENT RESULT ---
Name      : Rajesh
Roll Number: 101
Total Marks: 423
Percentage : 84.60%
Result     : PASS

===== STUDENT UTILITY PROGRAM =====
Enter your choice: 2

--- PART B: CONDITIONAL STATEMENTS ---
Enter your choice: 1
Enter a number: 24
24 is Even.

Enter your choice: 2
Enter first number: 25
Enter second number: 78
Enter third number: 46
Largest = 78

Enter your choice: 3
Enter percentage: 84.6
Grade: A

Enter your choice: 4
Enter day number (1-7): 3
Wednesday

===== STUDENT UTILITY PROGRAM =====
Enter your choice: 3

--- PART C: LOOPING STATEMENTS ---
Enter your choice: 1
Enter a number: 5
Multiplication Table of 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

Enter your choice: 2
Enter N: 10
1 2 3 4 5 6 7 8 9 10

Enter your choice: 3
Enter N: 10
Sum = 55

Enter your choice: 4
Enter number of terms: 8
Fibonacci Series: 0 1 1 2 3 5 8 13

===== STUDENT UTILITY PROGRAM =====
Enter your choice: 4

--- PART D: USER-DEFINED METHODS ---
Enter your choice: 1
Enter a number: 5
Factorial = 120

Enter your choice: 2
Enter a number: 17
17 is Prime.

Enter your choice: 3
Enter first number: 45
Enter second number: 72
```

Maximum = 72

Enter your choice: 4
Enter radius: 7
Area of Circle = 153.94

===== STUDENT UTILITY PROGRAM =====
Enter your choice: 5
Thank you for using Student Utility Program.

RESULT

Thus, the Student Utility Program was successfully developed in Java using basic programming constructs. The program demonstrates variables, data types, operators, input/output, conditional statements, loops, and parameterized user-defined methods with return values.